

# Probability Lecture 7

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## Conditional PMFs

We have done  $p_{X|A}(x|A) = P(X = x|A)$  now we want to condition on an another random variable, not an event

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

$$\sum_x p_{X|Y}(x|y) = 1$$

Also (similar to the multiplication rule for events)

$$p_{X,Y}(x, y) = p_Y(y)p_{X|Y}(x|y) = p_X(x)p_{Y|X}(y|x)$$

## Conditional PMFs involving more than two r.v.'s

$$p_{X|Y,Z}(x|y, z) = P(X = x|Y = y, Z = z) = \frac{P(X = x, Y = y, Z = z)}{P(Y = y, Z = z)} = \frac{p_{X,Y,Z}(x, y, z)}{p_{Y,Z}(y, z)}$$

Similar multiplication rule (for three events) applies

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_{Y|X}(y|x)p_{Z|X,Y}(z|x, y)$$

## Conditional Expectation

$$E[X|Y = y] = \sum_x xp_{X|Y}(x|y)$$

Expected Value Rule

$$E[g(x)|Y = y] = \sum_x g(x)p_{X|Y}(x|y)$$

## Total probability and expectation theorems

$A_1, \dots, A_n$  form a partition of  $\Omega$

$$p_X(x) = \sum_y p_Y(y)p_{X|Y}(x|y)$$

$$E[X] = \sum_y p_Y(y)E[X|Y = y]$$

## Independence

- **of two events:**  $P(A \cup B) = P(A)P(B)$   $P(A|B) = P(A)$
- **of a r.v. and an event:**  $P(X = x \text{ and } A) = P(X = x)P(A)$  for all  $x$   
or  $p_{X|A}(x) = p_X(x)$  for all  $x$   
or  $P(A|X = x) = P(A)$  for all  $x$
- **of two r.v.:**  $P(X = x \text{ and } Y = y) = P(X = x)P(Y = y)$  for all  $(x, y)$   
or  $p_{X,Y} = p_X(x)p_Y(y)$  for all  $x, y$   
or  $p_{X|Y}(x|y) = p_X(x)$

## Independence and Expectations

$$E[XY] = E[X]E[Y]$$

$$E[g(x, y)] \quad g(x, y) = xy$$

$$\sum_x \sum_y xyp_{X,Y}(x, y) = \sum_x \sum_y xyp_X(x)p_Y(y) = E[X]E[Y]$$

The same holds for  $g(X), h(Y)$

## Independence and Variance

If  $X, Y$  are independent  $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

## Variance of the binomial

$X$ : binomial with parameters  $n, p$  - number of successs in  $n$  independent trials

$X_i = 1$  if  $i$ th trial is a success

$X_i = 0$  otherwise

$X = X_1 + \cdots + X_n$

$$\text{var}(X) = \text{var}(X_1) + \cdots + \text{var}(X_n) = n\text{var}(X_1) = np(1 - p)$$